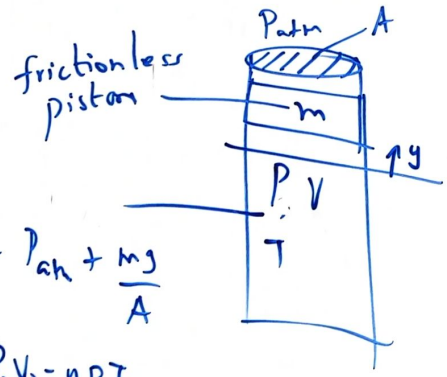
3-7-26

Waves Lec 7

Examples of physical systems executing SHM when "displaced" by a small value



In equil.

$P_0 = P_{atm} + \frac{mg}{A}$

$P_0 V_0 = nRT$

When the piston is displaced by y

$\Delta V = Ay$

$PV = nRT = \text{const}$

(assuming isothermal process)

$\Delta(PV) = 0$

$\Delta P\, V + P \Delta V = 0$

$P = -\frac{\Delta P}{\Delta V}$

$P = -\left[\frac{\Delta P}{(\Delta V/V)}\right]_T =$ isothermal Bulk modulus

$P = -V\left(\frac{\partial P}{\partial V}\right)_T \equiv B_T$

$\boxed{P = \frac{nRT}{V}}$

$\left(\frac{\partial P}{\partial V}\right)_T = -\frac{nRT}{V^2}$

$\left[-V\left(\frac{\partial P}{\partial V}\right)_T\right] = \frac{nRT}{V} = P$

$B_s = -V\left(\frac{\partial P}{\partial V}\right)_S$

$PV^\gamma = \text{const} = C$

$P = \frac{C}{V^\gamma} = CV^{-\gamma}$

$\left(\frac{\partial P}{\partial V}\right)_S = -C\gamma V^{-\gamma-1}$

$-V\left(\frac{\partial P}{\partial V}\right)_S = \gamma C V^{-\gamma}$

$\boxed{B_S = \gamma P}$

$V\left(\frac{\partial P}{\partial V}\right)_S$

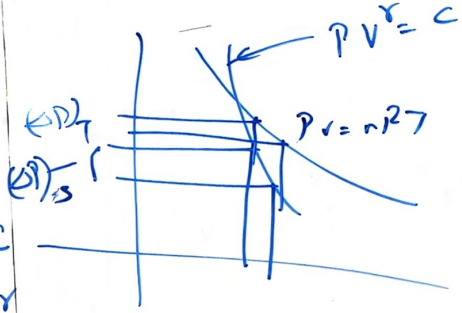
$P^\gamma = \text{const} = C$

$P = \frac{C}{V^\gamma} = CV^{-\gamma}$

$\left(\frac{\partial P}{\partial V}\right)_S = -C\gamma V^{-\gamma-1}$

$\left(\frac{\partial P}{\partial V}\right)_S = -\gamma C V^{-\gamma}$

$= \gamma P$



$B = -V\left(\frac{\partial P}{\partial V}\right)_S$

$= -V\left(\frac{\Delta P}{\Delta V}\right)_S$

$\Delta P = -B_s\left(\frac{\Delta V}{V_0}\right)$

$F = \Delta P A = -\frac{A B_s A y}{V_0 \to A L_0}$

$F = -\frac{B_s A y}{L_0} = -m\frac{d^2y}{dt^2}$

$\frac{d^2y}{dt^2} = -\frac{B_s A}{m L_0} y$

$\omega^2 = \frac{B_s A}{m L_0}$

$\omega = \sqrt{\frac{\gamma P_0 A}{m L_0}}$

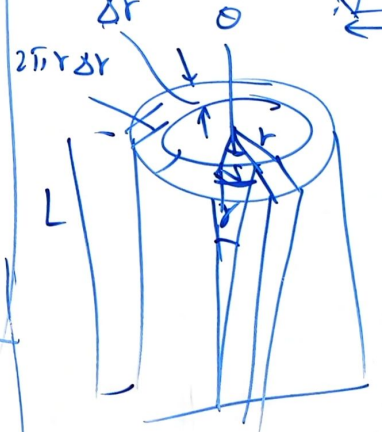
$\tau = \frac{G\theta}{L} 2\pi \int_0^R r^3 dr$

$\tau = -\frac{\pi G R^4}{2L}\theta$

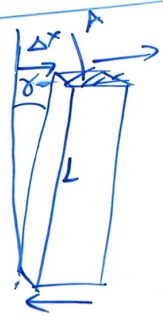
Torsional oscillations

$\tau = I\frac{d^2\theta}{dt^2} = -C\theta$

$\omega = \sqrt{C/I}$



$\tau = -C\theta$



$\sigma_{shear} = \Delta F / A$

γ = shear stress

$= \Delta x / L$

$\frac{\sigma_{shear}}{\gamma} = G$ rigidity modulus.

$\Delta F = G\gamma\Delta A$

$\gamma = \frac{s}{L} = \frac{r\theta}{L}$

$\delta F = G\frac{r\theta}{L} \cdot 2\pi r \delta r$

(torque) $\delta\tau = r\delta F = \frac{G\theta}{L} 2\pi r^3 dr$